

# Rethinking Efficient Decoder Design: A Systematic Characterization of Where Decoder Computation Helps

Anonymous WACV Datasets Track submission

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## Abstract

*The ABSTRACT is to be in fully justified italicized text, at the top of the left-hand column, below the author and affiliation information. Use the word “Abstract” as the title, in 12-point Times, boldface type, centered relative to the column, initially capitalized. The abstract is to be in 10-point, single-spaced type. Leave two blank lines after the Abstract, then begin the main text. Look at previous WACV abstracts to get a feel for style and length.*

Table 1. Predeclared backbone  $\times$  dataset grid. Each cell is a matched full/EffiDec3D decoder pair sharing one encoder, run through the full metric suite. \*: completed; remaining cells in progress.

| Backbone            | BTCV<br>(CT) | FeTA<br>(MRI) | MSD-BrTu<br>(MRI) | MSD-Lung/HV<br>(CT) |
|---------------------|--------------|---------------|-------------------|---------------------|
| 3D UX-Net (CNN)     | ✓*           | ✓             | ✓                 | ✓                   |
| Swin UNETR (Trans.) | ✓            | ✓             | ✓                 | ✓                   |
| SwinUNETR-V2 (opt.) | ✓            | ✓             | ✓                 | ✓                   |
| MedNeXt-M-K3 (opt.) | ✓            | ✓             | ✓                 | ✓                   |

## 1. Experimental Setup

**Experiment plan.** Our study reproduces EfficDec3D [14] and evaluates it on a full backbone  $\times$  dataset grid (Table 1). For each backbone that EfficDec3D actually equips with an efficient decoder—3D UX-Net [11], Swin UNETR [16], and optionally SwinUNETR-V2 [8] and MedNeXt [15]—we pair a full and an EfficDec3D decoder and run that matched pair on each of three to four representative datasets from the EfficDec3D evaluation: BTCV abdominal CT, FeTA [13] fetal-brain MRI, and predeclared MSD tasks [1] (Task01 BrainTumour, and Task06 Lung or Task08 HepaticVessel). Every grid cell is therefore a matched full/efficient pair on which the full metric suite is computed; cross-dataset consistency is read across each row and cross-backbone consistency down each column. UNETR [7] and nnU-Net [10] are excluded because EfficDec3D builds no decoder pair for them, so a matched flip comparison is undefined.

**Dataset and split.** The primary task is the 13-organ BTCV/Synapse abdominal CT benchmark with 18 training and 12 validation volumes: spleen, right and left kidney, gallbladder, esophagus, liver, stomach, aorta, inferior vena cava, portal and splenic veins, pancreas, and right and left adrenal glands. Images are intensity-windowed and resampled following the public EfficDec3D pipeline; the same validation subjects are paired across all model comparisons on

a given dataset.

**Matched models.** Within each cell the two models share an identical encoder and differ *only* in the decoder: the full decoder versus the EfficDec3D decoder (channels reduced to the minimum encoder width, additive skip aggregation, resolution factor two). Holding the encoder fixed makes any prediction difference attributable to decoder capacity alone; consequently the study characterizes the decoder and draws no conclusions about the encoder.

**Implementation.** We reproduce the EfficDec3D training protocol on a single NVIDIA RTX 5090 (32 GB, Blackwell) under PyTorch 2.6 / CUDA 12.8 with MONAI, using BF16 mixed precision (no loss scaling). Each model is trained for 45,000 optimizer steps (validation every 250) with AdamW at learning rate  $10^{-3}$ , a Dice-cross-entropy loss, four  $96^3$  random-crop samples per volume, and full in-memory caching. Sliding-window inference uses a  $96^3$  window, batch size four, overlap 0.7, and constant blending. We evaluate one full run and two independently initialized EfficDec3D runs per cell to expose seed sensitivity. Segmentation quality is reported with Dice and 95th-percentile Hausdorff distance (HD95), and efficiency with parameters, multiply-accumulate operations (MACs), latency, and peak memory.

**Core quantities.** Let  $\hat{y}_E(v)$ ,  $\hat{y}_F(v)$ , and  $y(v)$  be the efficient prediction, full prediction, and ground-truth label at voxel  $v$ . We measure where the full decoder changes predictions with *prediction flips*, the standard tool for comparing two classifiers on paired data [2, 17]:

$$P(v) = \mathbb{I}[\hat{y}_E(v) \neq y(v) \wedge \hat{y}_F(v) = y(v)], \quad (1)$$

$$N(v) = \mathbb{I}[\hat{y}_E(v) = y(v) \wedge \hat{y}_F(v) \neq y(v)], \quad (2)$$

$$U(v) = P(v) - N(v), \quad (3)$$

a *positive* flip (full corrects efficient), a *negative* flip (full degrades efficient), and the *net* flip.  $P$  and  $N$  are exactly the discordant cells of the paired-agreement table behind McNemar’s classifier-comparison test [3, 12]; volume rates are  $R_{\text{pos}} = \text{mean}_v P(v)$ ,  $R_{\text{neg}} = \text{mean}_v N(v)$ , and  $R_{\text{net}} = R_{\text{pos}} - R_{\text{neg}}$ . As test-time signals computable from the efficient model alone we use predictive entropy  $H(v) = -\sum_c p_c(v) \log p_c(v)$ , confidence  $\max_c p_c(v)$ , and Monte-Carlo dropout variance [4].

**Characterization metrics.** We group the analysis into three questions; each metric is computed per grid cell and aggregated over subjects with a bootstrap, since the twelve validation volumes—not the correlated voxels—are the independent units.

(1) *Heterogeneity: is the benefit evenly distributed?*

- **O1, boundary error ratio.** Mean voxel error on the eroded boundary band divided by the eroded-interior error;  $>1$  indicates boundary-concentrated error.
- **O4, per-organ difficulty.** Per-class Dice and mean entropy  $H$ , ranking the structures the decoder struggles on.
- **O10, size–difficulty relation.** Spearman  $\rho$  between organ volume and difficulty, and the partial correlation of  $H$  and error given log-volume.

(2) *Concentration: does a small subset carry most net benefit?*

- **O2, entropy sparsity.**  $\Pr[H(v) > \tau]$ , the fraction of high-entropy voxels.
- **O5, uncertainty–benefit correlation.** Subject-level Pearson  $r(H, U)$  between binned entropy and net flips.
- **O9<sup>orc</sup>, voxel recovery.**  $\sum_{\text{top-}k\%} P(v) / \sum P(v)$ , the share of positive flips in the highest-entropy  $k\%$  of voxels—a non-deployable upper bound.
- **O6, training persistence.** Mean entropy across checkpoints, testing whether concentration is a training transient.

(3) *Predictability: can a test-time signal find the regions?*

- **O3, flip discrimination and calibration.** AUROC and AUPRC of entropy predicting flips [9], and expected calibration error (ECE) [6].
- **O9, opportunity curve.** A risk–coverage curve [5] over contiguous  $16^3$  blocks (plus a 4-voxel halo) ranked

Table 2. Reproduction fidelity of the matched 3D UX-Net pair, per dataset. We report mean Dice of our runs against the published EffiDec3D value;  $\Delta$  is our Effi-UX minus the published Effi-UX. Data were reconstructed with identity affine metadata, so this is a controlled reimplement, not an exact reproduction.

| Dataset                  | Full-UX | Effi-UX | pub. Effi-UX | $\Delta$ |
|--------------------------|---------|---------|--------------|----------|
| BTCV (CT, 13-organ)      | .792    | .773    | .793         | −.001    |
| FeTA (MRI)               | —       | —       | —            | —        |
| MSD BrainTumour          | —       | —       | —            | —        |
| MSD Lung / HepaticVessel | —       | —       | —            | —        |

[TODO: Fill FeTA/MSD rows as cells complete.]

by entropy, reporting net-flip utility recovered versus a matched-random selector.

- **O11, routing-signal comparison.** Subject-level correlation of entropy, confidence, and MC-dropout with net flips.

Cross-dataset (O7) and cross-backbone (O8) consistency re-evaluate these metrics across the rows and columns of Table 1.

[TODO: Completed results below are the 3D UX-Net/BTCV grid cell. Remaining cells (other backbones and datasets) are in progress; cross-backbone (columns) and cross-dataset (rows) consistency are reported once enough cells are filled.]

## 2. Results

### 2.1. Controlled Reimplementation

We reproduce the matched Full-UX/Effi-UX pair and report its fidelity per dataset (Table 2). On BTCV our Effi-UX reaches 0.773 mean Dice over two seeds, 1.7–2.2 points below the published 0.793, while Full-UX is within 0.6 points of its published value; the residual gap reflects our reconstructed identity-affine data rather than the paper’s processed set, and a FP32-versus-BF16 check ruled out a numerical cause. Efficiency reproduces the published reductions: Effi-UX uses  $14.1\times$  fewer MACs,  $17.9\times$  fewer parameters,  $\sim 5\times$  lower latency, and  $\sim 7.7\times$  lower peak memory. Treating these matched models as fixed, the rest of the paper characterizes *where* the full decoder’s extra computation changes predictions—the question EffiDec3D’s aggregate evaluation leaves open.

### 2.2. Heterogeneity Metrics

The full decoder’s benefit is small and *bidirectional*: Full-UX corrects 0.290% of Effi-UX’s voxels ( $R_{\text{pos}}$ ) while degrading 0.225% ( $R_{\text{neg}}$ ) for seed 1, and 0.339%/0.209% for seed 0, so the net flip rate  $R_{\text{net}}$  is only 0.065–0.130% of voxels (Table 3). Reporting positive flips alone would overstate the decoder’s benefit roughly twofold.

Table 3. Prediction-flip rates across efficient-decoder seeds (fractions of evaluated voxels). Positive/negative flips are the discordant cells of the paired comparison [12, 17].

| Metric                              | seed 0                                          | seed 1                                          |
|-------------------------------------|-------------------------------------------------|-------------------------------------------------|
| Positive flip rate $R_{\text{pos}}$ | .00339                                          | .00290                                          |
| Negative flip rate $R_{\text{neg}}$ | .00209                                          | .00225                                          |
| Net flip rate $R_{\text{net}}$      | .00130                                          | .00065                                          |
| Net flip rate, subject 95% CI       | <span style="color: red;">[TODO: re-run]</span> | <span style="color: red;">[TODO: re-run]</span> |

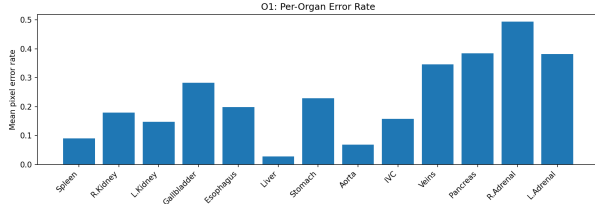


Figure 1. Per-organ error rate for Effi-UX (seed 1). Difficulty is strongly uneven across anatomy.

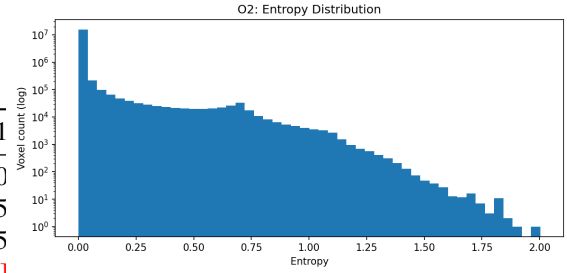
This small net benefit is far from uniformly distributed. Error concentrates at structure boundaries: mean boundary error is 0.1892 versus 0.0481 in the eroded interior, a  $3.93\times$  ratio [TODO: add subject 95% CI from re-run]. It is also organ-dependent (Fig. 1), with the adrenal glands, pancreas, and veins hardest, and it scales with structure size: the organ-size/difficulty Spearman correlation is  $\rho = -0.54$ , and entropy still explains organ difficulty after controlling for log volume (partial  $r = 0.81$ ). Decoder benefit is therefore heterogeneous across region, organ, and scale.

## 2.3. Concentration Metrics

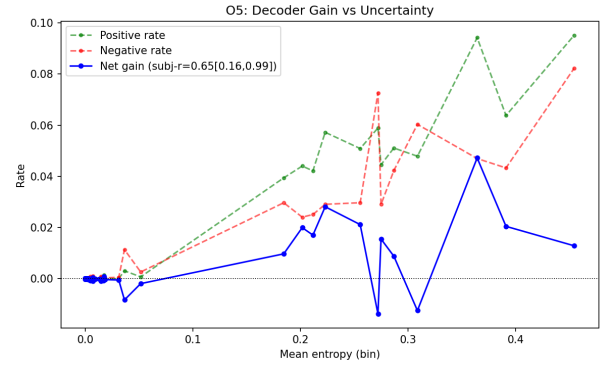
A small, high-uncertainty subset carries most of the net benefit. Predictive entropy is spatially sparse: the median voxel entropy is near zero and only 1.18% of voxels exceed 0.5 (Fig. 2a). Net flips concentrate in exactly this tail—the subject-level Pearson correlation between binned entropy and net flips is  $r = 0.646$  (95% CI [0.158, 0.994]) for seed 1 and  $r = 0.665$  ([0.171, 0.996]) for seed 0, both bounded above zero (Fig. 2b). A voxel-level oracle makes the concentration explicit: ranking foreground voxels by entropy, the top 20% contain 86.4% of all positive flips in both seeds. This concentration is not a transient of early training: mean entropy falls from 0.0279 at 5k iterations to 0.0104 at 45k, yet the sparse difficult residual persists.

## 2.4. Predictability Metrics

Concentration is only actionable if the beneficial regions can be found without the ground truth. We assess this as an error-discrimination problem: predictive entropy should separate flip voxels from the rest [9]. Per-subject AUROC



(a) Voxel entropy (log-count).



(b) Net flips vs. entropy.

Figure 2. Concentration of difficulty and net flips (seed 1). Uncertainty is spatially sparse (a), and net flips rise with it (b).

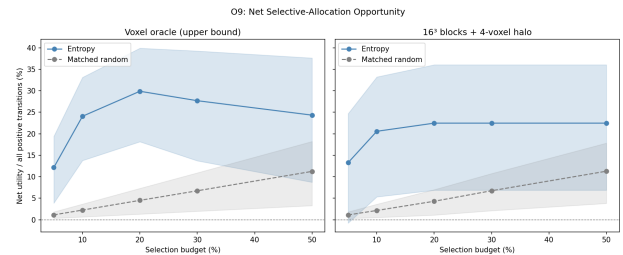


Figure 3. Region-level opportunity (seed 1). Contiguous  $16^3$  blocks ranked by entropy recover far more net flip utility than a matched-random selector at every budget  $\geq 10\%$ ; the left panel is a non-deployable voxel oracle.

and AUPRC of entropy predicting error, together with confidence calibration (ECE) [6], are reported with a subject bootstrap [TODO: fill AUROC/AUPRC/ECE + CI from re-run; the pooled-bin entropy-error correlation  $r = 0.97$  is retained only as a descriptive diagnostic].

We then convert this into a deployable, selective-prediction analysis [5]. Non-overlapping  $16^3$  blocks are ranked by mean entropy and expanded by a 4-voxel halo; we measure net flip utility recovered against a matched-random selector under a paired subject bootstrap (Fig. 3). At the predeclared 20% block budget—executing 29.1% of the foreground-union volume after halo expansion—

entropy-ranked regions attain a net utility of 0.225 versus 0.043 for random, a roughly fivefold gain, with a paired 95% lower bound of  $0.056 > 0$ ; the advantage holds at every budget  $\geq 10\%$  and vanishes only at 5%. These regions capture all positive flips but also 77.5% of the negative ones, so the *net* gain, not positive recovery, is the honest headline.

Finally, we compare cheap routing signals against held-out net flips per subject. Entropy and confidence are statistically indistinguishable [TODO: fill subject-level correlations + paired CI from re-run; pooled values are 0.655 and 0.663], and Monte-Carlo dropout is invalid for this checkpoint because the architecture has no active dropout (its stochastic variance is effectively zero). We therefore retain confidence as a required baseline and do not claim entropy is uniquely optimal.

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